Paper 1: Jiang et al. (2024)

Title: Research on the Identification and Classification of Marine Debris Based on Improved YOLOv8

What the paper was about

This paper looks at how deep learning can be used to detect and classify rubbish underwater. Detecting debris is difficult because underwater images often have poor lighting, blurry objects and particles floating in the water. Marine animals and plants can also look similar to rubbish, making it harder for detection models to tell them apart. The authors try to improve the YOLOv8 object detection model so it can identify different types of marine debris more accurately. The goal is to make underwater debris detection more reliable for autonomous underwater robots that could eventually help with cleaning up the ocean.

Method used

The authors use an improved version of YOLOv8, which is a deep learning model that detects objects and draws bounding boxes around them. They make three main changes to the original model. First, they add a CloFormer module to help the network capture both fine details and surrounding information in underwater images. Second, they introduce a C2f-SCConv module, which reduces unnecessary computations and helps the model extract useful features more effectively. Finally, they replace the original bounding box loss function with WIoU v3, which helps the model learn from different-quality training examples and improve object localisation. The improved model is then trained and tested on underwater debris images to see whether these changes make a difference.

Benefits

The biggest advantage is that the improved YOLOv8 performs better than the original model at recognising underwater rubbish. On the TrashCan-Instance dataset, the authors report a 5.7 percentage-point improvement in mAP@0.5 and a 6 percentage-point improvement in F1 score. On TrashCan-Material, the improvements are 5.5 and 5 percentage points respectively. This shows that the changes help the model detect debris more accurately across different object and material categories. The model also uses information from both the objects and their surroundings, which is useful when rubbish blends into the seabed or appears unclear because of underwater conditions. Since YOLO is designed for efficient object detection, the approach is also relevant to underwater robots that need to process camera images quickly.

Dataset used

The authors use the publicly available TrashCan dataset, which contains 7,212 annotated underwater images collected from JAMSTEC's deep-sea video database. The

images include different types of rubbish, marine animals, plants and parts of underwater robots. The dataset has two versions: TrashCan-Instance, which labels rubbish by object type, such as bottles, bags and cans, and TrashCan-Material, which labels rubbish by material, such as plastic, metal and rubber. Both versions contain bounding box annotations that can be used to train object detection models. This is especially relevant to our project because we are also using the TrashCan dataset to train and compare deep learning models.

Limitations

- Underwater visibility: Even with the improvements, detecting small or partially hidden objects remains difficult when images are dark, blurry or affected by suspended particles.
- Dataset diversity: The experiments mainly use TrashCan, so it is unclear how well the model would perform in completely different underwater environments or with images from different cameras.
- Computational complexity: Adding extra modules makes the architecture more complicated than standard YOLOv8, which may affect its suitability for underwater robots with limited computing resources.
- Object detection only: The model predicts bounding boxes rather than detailed segmentation masks, so it cannot identify the exact shape of each debris object.
- Real-world testing: The results are based on an existing image dataset rather than a full autonomous underwater cleanup operation, so its performance during actual deployment still needs further testing.

Our project is similar because we also use TrashCan to detect and classify marine debris. However, instead of improving YOLOv8, we focus on Faster R-CNN and RetinaNet, comparing different detection architectures and techniques for improving performance on underwater images.

Paper 2: Đuraš et al. (2024)

Title: A Dataset for Detection and Segmentation of Underwater Marine Debris in Shallow Waters

What the paper was about

This paper focuses on improving underwater marine debris detection by creating a dataset that better represents real underwater conditions. Most existing datasets, like TrashCan, contain images collected from deep-sea environments, but underwater rubbish is also found in shallow coastal waters where lighting, water clarity and backgrounds can vary a lot. These differences make it difficult for deep learning models to detect rubbish accurately in new locations. The authors introduce the SeaClear Marine Debris Dataset, which contains images collected from different shallow-water environments. They also test deep learning models on the dataset to see how well they can detect debris and whether they can still perform accurately when the camera or underwater location changes.

Method used

The authors use remotely operated underwater vehicles (ROVs) equipped with cameras to collect images from different coastal locations. The images are manually annotated so that rubbish, marine life and robot parts can be identified. They then train two object detection models, Faster R-CNN and YOLOv6, using the dataset. Faster R-CNN works in two stages: it first identifies possible object locations and then classifies them, while YOLOv6 detects and classifies objects in a single stage, making it faster. Both models are initially pretrained on COCO and then fine-tuned on SeaClear. The authors also test an underwater image enhancement method that adjusts colour and contrast to see whether clearer images improve detection accuracy. Finally, they compare performance across different cameras and locations to understand how well the models generalise to new underwater environments.

Benefits

One of the biggest benefits is that the paper provides a publicly available dataset containing underwater images from several locations, rather than relying on a single environment. This makes it useful for training and testing models under more realistic conditions. The authors also show that YOLOv6 performs better than Faster R-CNN in their standard dataset split, achieving approximately 7 percentage points higher detection performance. YOLOv6 is also smaller and more suitable for real-time applications, which could be useful for underwater robots. Another important finding is that improving the visual quality of underwater images does not necessarily improve object detection accuracy. This helps researchers understand that simply adjusting

brightness and colour may not be enough to make detection models more reliable in different environments.

Dataset used

The authors create their own dataset called the SeaClear Marine Debris Dataset, which contains 8,610 underwater images collected using ROVs at five locations in Croatia and France. The images were captured using different cameras and under varying conditions, including clear water, murky water, natural lighting and areas with partially hidden rubbish. The dataset originally includes 40 object categories covering debris, marine animals, plants and underwater robot parts. Objects are annotated with bounding boxes and segmentation masks, allowing the dataset to be used for both object detection and instance segmentation. For their detection experiments, the authors combine some categories with fewer than 50 annotations, reducing the number of categories to 34. Unlike TrashCan, which mainly contains deep-sea images, SeaClear focuses on shallow-water environments.

Limitations

- Generalisation: The models perform well when training and testing images come from similar environments, but their accuracy drops when tested on different locations or cameras. This shows that they still struggle to recognise debris in unfamiliar conditions.
- Class imbalance: Some types of rubbish appear much more frequently than others. The authors have to combine categories with fewer than 50 annotations, which reduces the detail of the classification.
- Image enhancement: Improving the colour and contrast of underwater images does not consistently improve detection performance, so image enhancement alone cannot solve the problem.
- Limited locations: Although the dataset includes different environments, all the images come from five locations in Croatia and France. It may not represent underwater conditions in other parts of the world.
- Model performance: Faster R-CNN and YOLOv6 are tested as baseline models rather than extensively improved architectures, so there is still room to explore more advanced detection methods.

Our project also aims to detect marine debris automatically using deep learning, but we use the TrashCan dataset instead of SeaClear. This paper is useful because it shows how important dataset diversity and class imbalance are when training underwater detection models. It also helps explain why our project uses underwater image augmentation and repeat-factor sampling to improve detection performance.